EE4705 Project 2 · Group 9

# Collision-free path planning with STOMP

Kinova Gen3 and Franka Emika Panda, in MATLAB

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# Five tasks, one code base

| Task | What we did | Result |
|---|---|---|
| 1 | Completed STOMP on the Kinova Gen3 | Collision-free in 23 iterations |
| 2 | Moved to the Franka Emika Panda | Collision-free in 13 iterations |
| 3 | Product-of-Exponentials kinematics | Same path, 14× faster iterations |
| 4 | Three custom multi-obstacle scenes | 30 of 30 seeded runs collision-free |
| 5 | Kept the gripper’s y-axis upright | Maximum tilt 36.8° → 1.9° |

Every result is checked against the robot meshes, at the waypoints and between them.

# STOMP: sample, score, weight, update

### 1 Sample

20 noisy trajectories, smooth noise from $R^{-1}$

### 2 Score

Obstacle cost at every waypoint

### 3 Weight

Cheaper samples get higher probability

### 4 Update

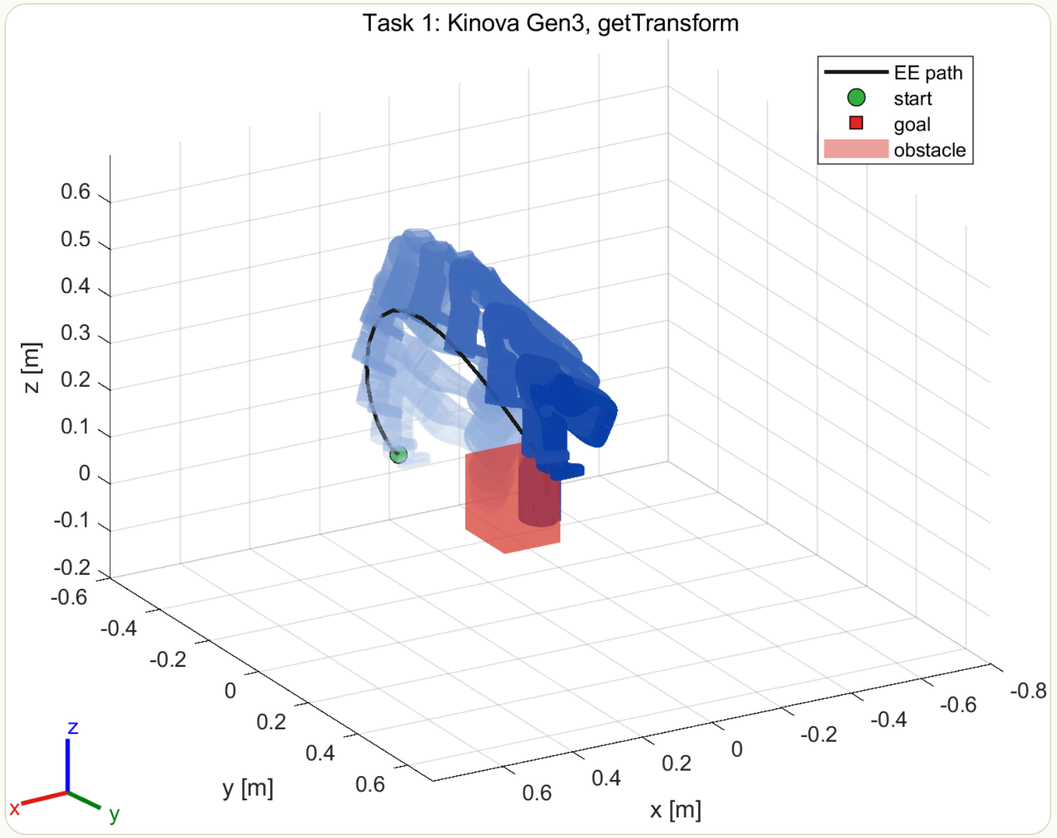
Move θ by the weighted noise, smoothed by M

**Obstacle cost, summed over the collision spheres**

$$q = \Sigma \max(\varepsilon + r - d(x), 0) \cdot \|\dot{x}\|$$

d comes from a signed distance field of a 2 cm voxel world. No gradient of the cost is needed.

# Task 1: Kinova Gen3 around the box



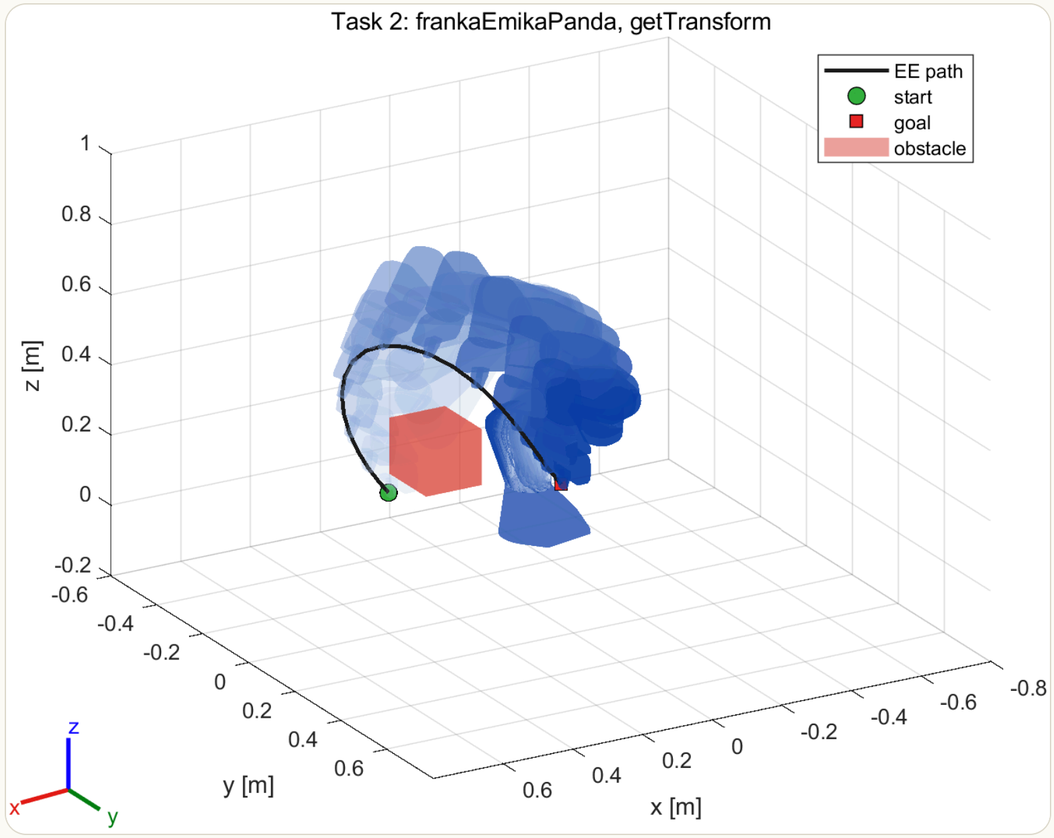

# 23 iterations

8 of 20 waypoints collided at the start; none at the end

## Bugs we fixed in the sample code

- Samples leaving the world scored zero cost
- Control cost added cross-joint terms, 45% too high
- The last ~6 cm of the arm had no spheres

# Task 2: Franka Emika Panda




## A harder arm

7 joints with tight limits, link offsets, a two-finger gripper

## Tool point at the fingertips

Fingers frozen; a TCP frame added 0.1034 m past the flange

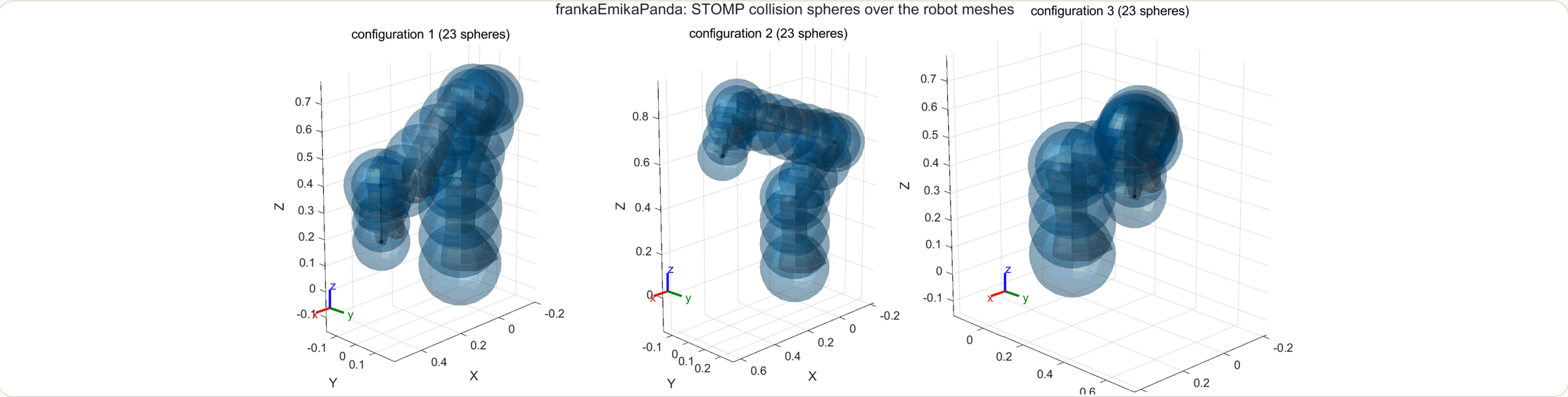
## Random restarts for IK

Without them the goal was missed by 8.9 cm

## A scene that really collides

Box on the joint-space path: 7 of 20 waypoints; solved in 13 iterations

# Collision spheres fitted to the meshes




**5 → 10–15 cm**
sphere radius, one per link

**99%**
of mesh vertices inside the spheres

**5 → 2 cm**
safety margin, now a real margin

# One small cost change is not convergence

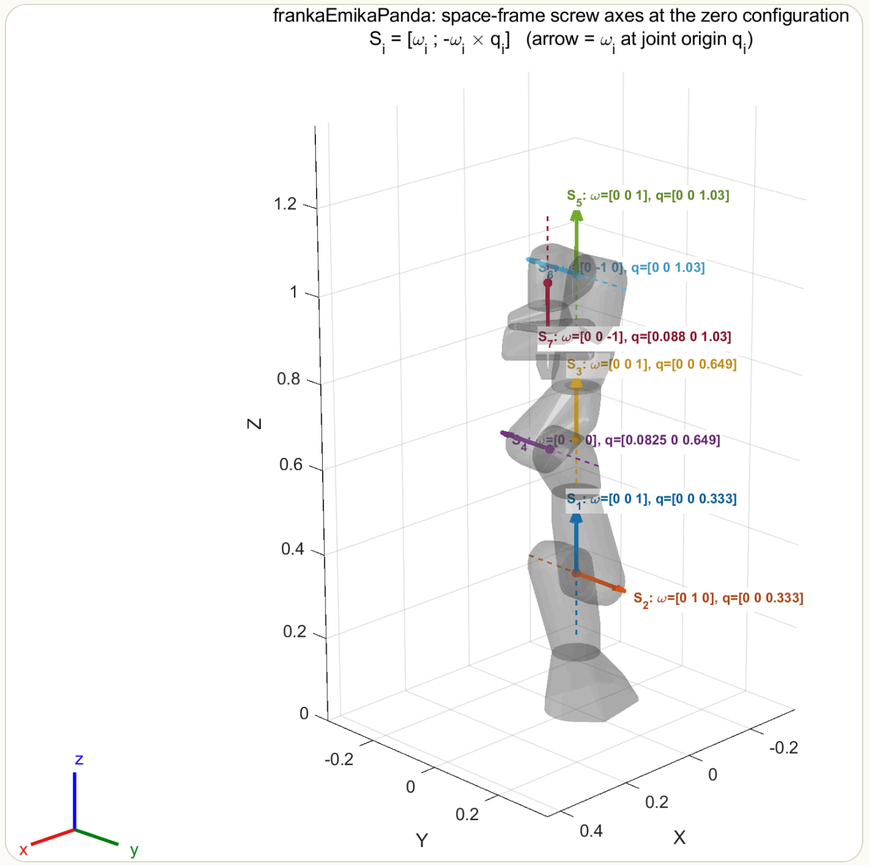
**29 → 35**

collision-free runs out of 35, going from the original stop rule to three consecutive small changes

| Scene | P = 1 | P = 3 |
|---|---|---|
| Task 4 S3 | 1 / 5 | 5 / 5 |
| Task 4 S1 | 4 / 5 | 5 / 5 |
| Task 5 constrained | 4 / 5 | 5 / 5 |
| Other four scenes | 5 / 5 | 5 / 5 |

The price is more iterations: 9% more at best, about 6× in the constrained case.

# Task 3: screw axes from the robot model




$$T(\theta) = e^{[S_1]\theta_1} \cdots e^{[S_k]\theta_k} M$$

## ω: joint axis in the base frame

The URDF axis is in the joint's own frame, so it must be rotated

## v = −ω × q

q is the joint origin at the zero configuration

## Checked with the matrix log

Both methods agree to 4e-16; computed once per robot

# Task 3: the same answer, much faster

**1e-15**

largest difference from getTransform, 500 random poses

**35×**

faster forward kinematics for all frames (Panda)

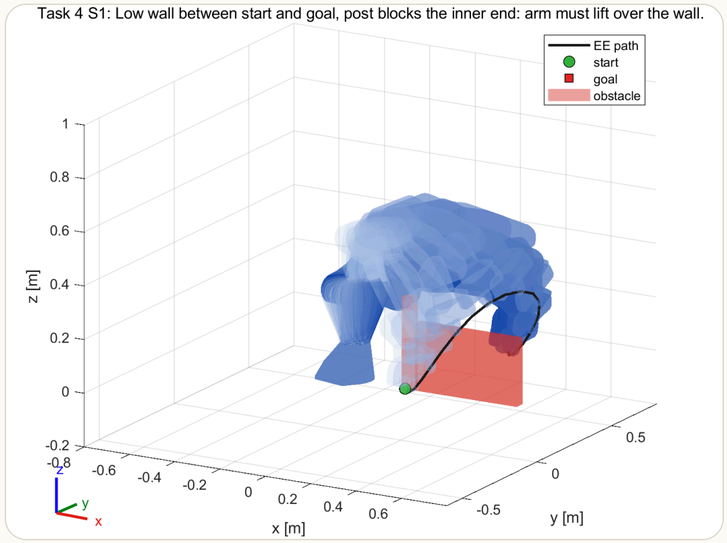
**14×**

faster STOMP iterations: 527 ms → 38 ms

**Why it is fast:** one running product gives every frame, with closed-form exponentials instead of expm.

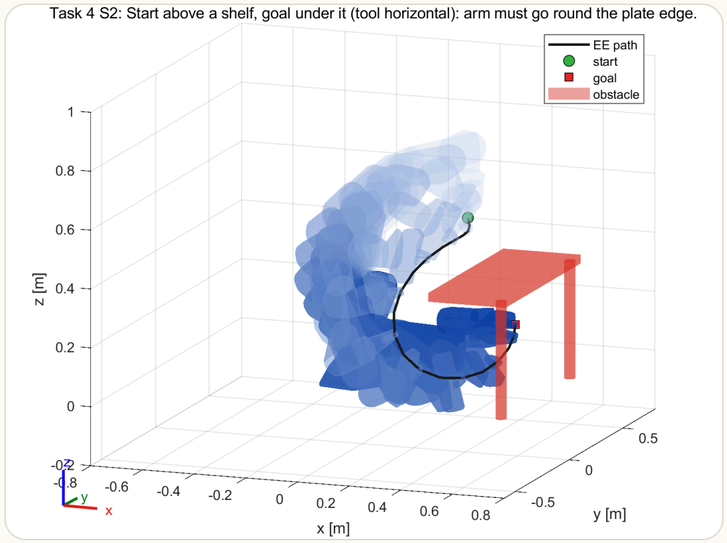
**Same seed, same plan:** PoE and getTransform give an identical trajectory.

# Task 4: three scenes that force a detour



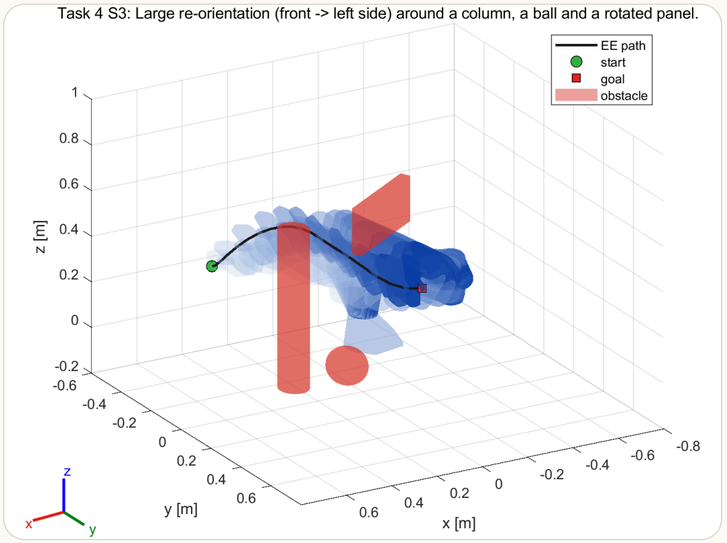

## Over the wall

10 of 20 waypoints collide at the start; 32 iterations




## Under the shelf

6 of 20 waypoints collide at the start; 20 iterations

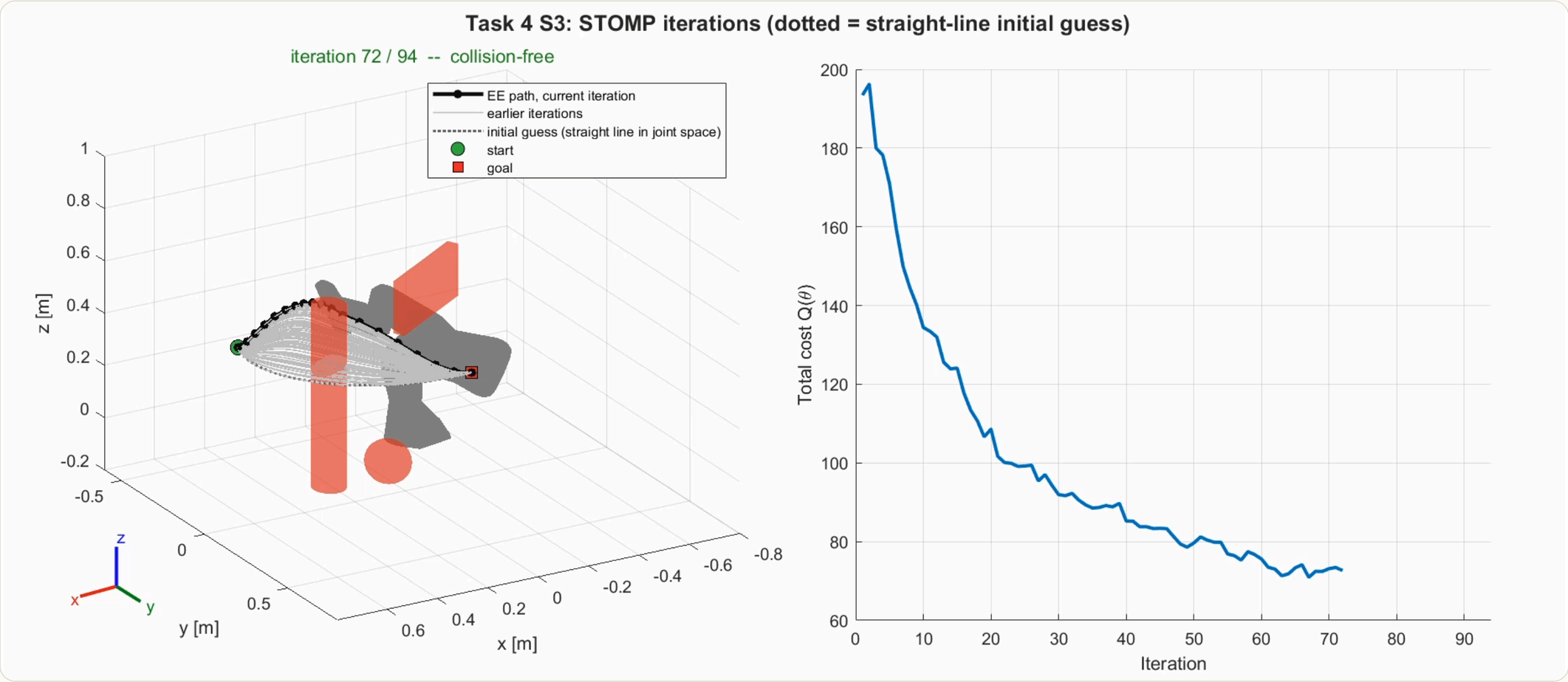



## Around the column

5 of 20 waypoints collide at the start; 94 iterations

**10 of 10 seeds collision-free in every scene.** Boxes, rotated boxes, spheres and cylinders in one voxel world.

# STOMP pushing the path off the column



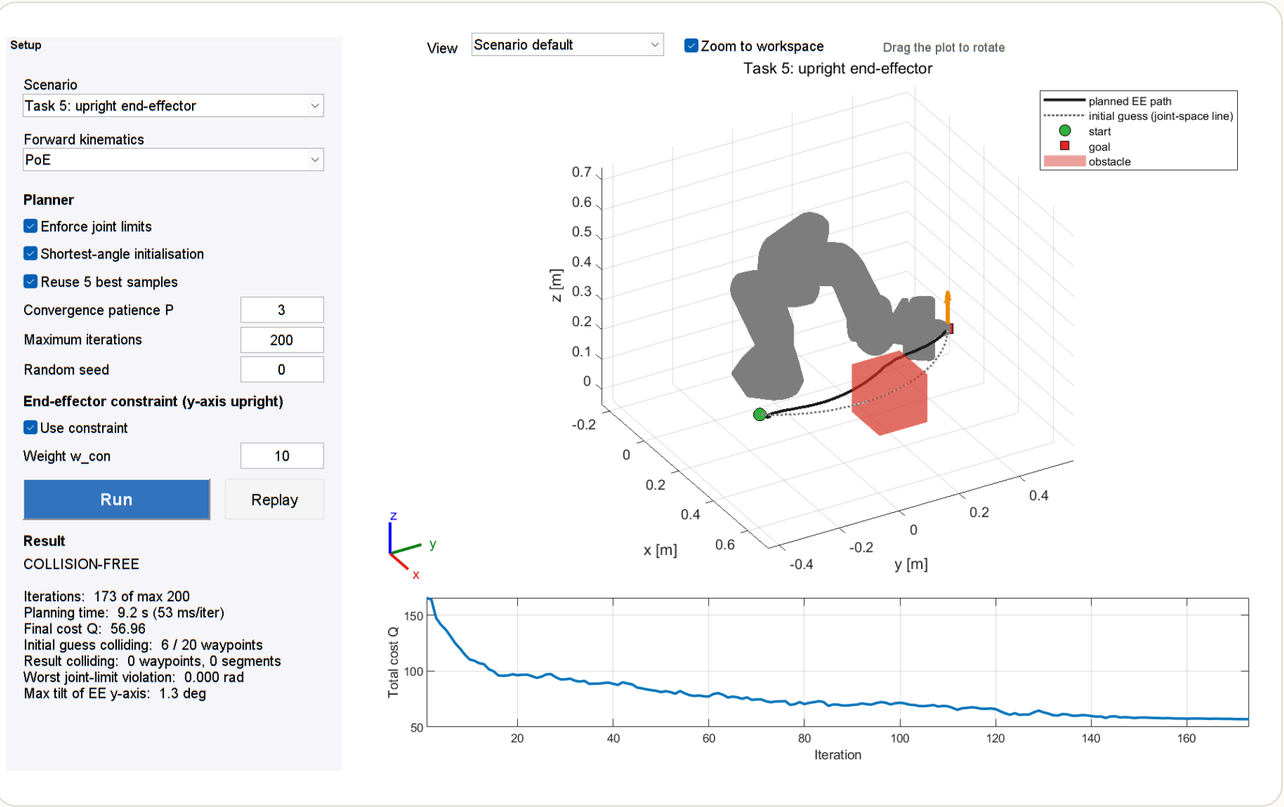
# Extending STOMP: joint limits and sample reuse

| Variant | Collision-free | Worst limit violation | S3 iterations |
|---|---|---|---|
| No extensions | 35 / 35 | 0.52 rad | 79 |
| Joint limits | 31 / 35 | 0 | 147 |
| Limits + sample reuse (our default) | 35 / 35 | 0 | 85 |
| Limits + noise decay | 29 / 35 | 0 | 78 |

**Limits matter:** plain STOMP left the Panda's limits by up to 30°, which a real arm cannot execute.

**Reuse is free:** the 5 cheapest samples are kept, with no extra kinematics. Noise decay made things worse, so it stays off.

# Practical features




## GUI

Pick a scene and options, plan, and rotate the view

## ±π joint angles

Shortest-angle start cuts a demo’s end-effector path from 3.31 m to 0.79 m

## Ground truth

Mesh check at waypoints and 9 steps between each pair

# Task 5: carrying a cup of coffee

**Constraint cost at each waypoint (handout option 2)**

$$q = \|\hat{y} - \hat{z}\|_1$$

$\hat{y}$ is the gripper's y-axis in the world frame, $\hat{z}$ is world up. The cost is zero only when $\hat{y}$ points straight up.

### Cheap to compute

$\hat{y}$ is a column of the rotation matrix FK already gives

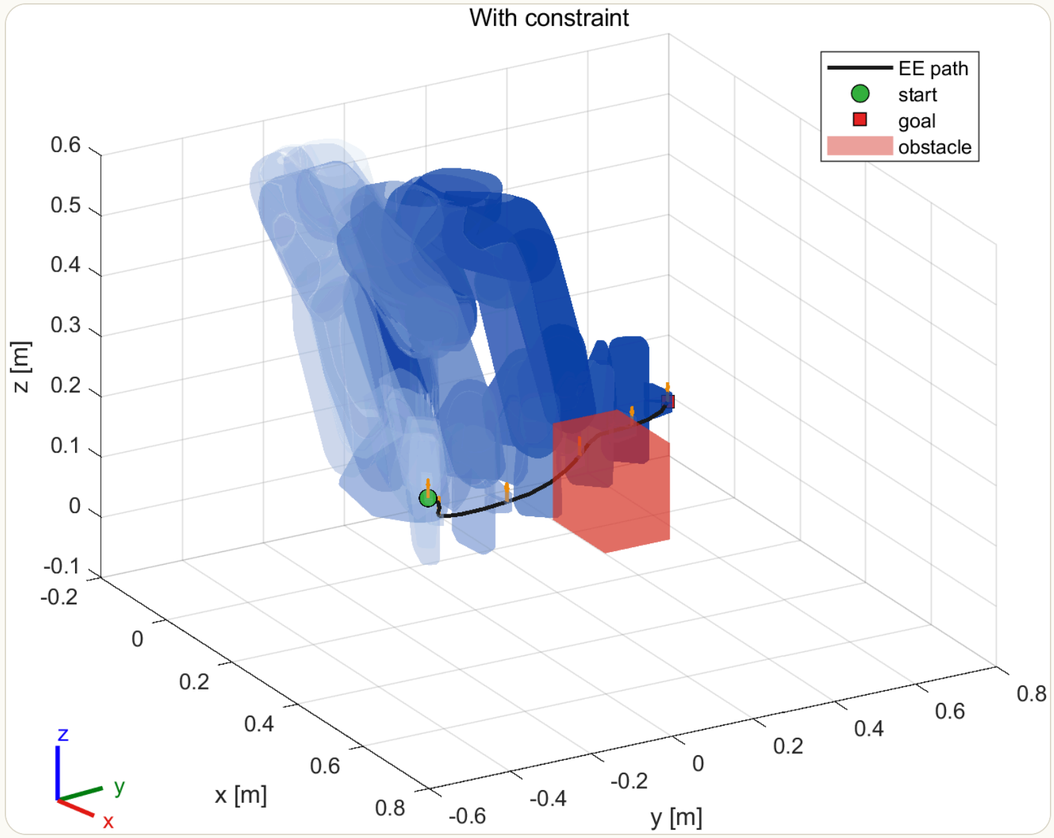
### Added to S

Weighted by w_con, so upright samples get higher probability

### One axis only

The gripper may still spin about the vertical; the cup stays upright

# Task 5: designing an upright scene




## Upright at both ends

0.5 m from the base axis, tool horizontal, y-axis vertical

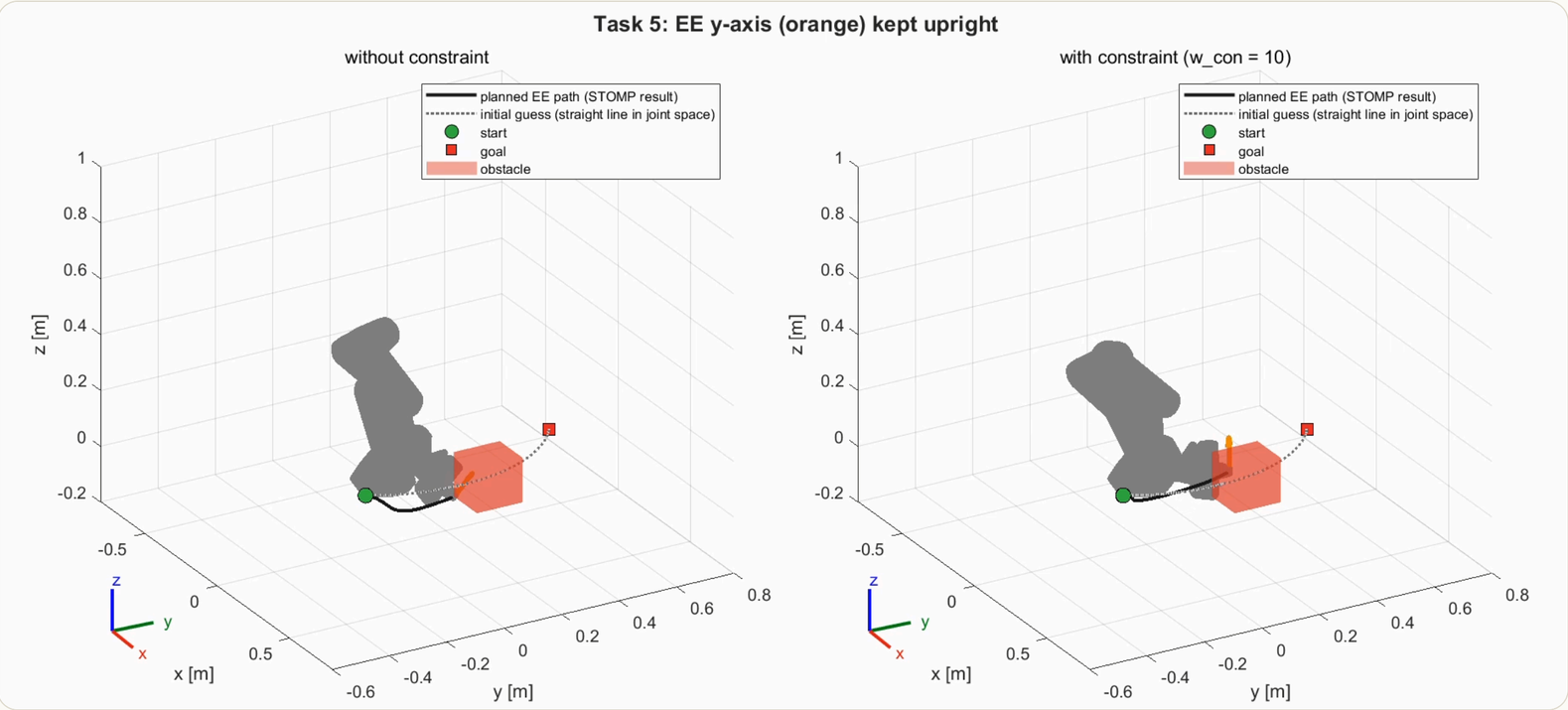
## A box on the way

16 cm cube on the arc: 6 of 20 waypoints collide at the start

## Height set by reachability

At 0.30 m IK stopped 2.8 cm off, joint 5 at its limit; at 0.20 m it is exact

# Task 5: with and without the constraint



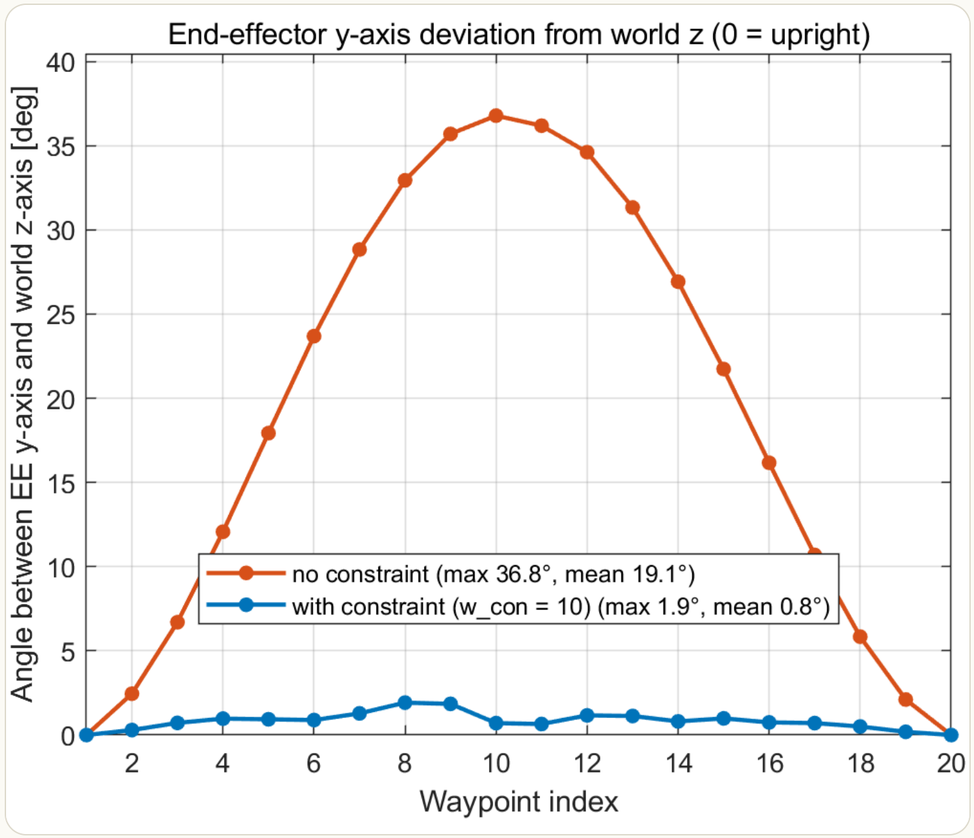

**36.8° → 1.9°**
maximum tilt

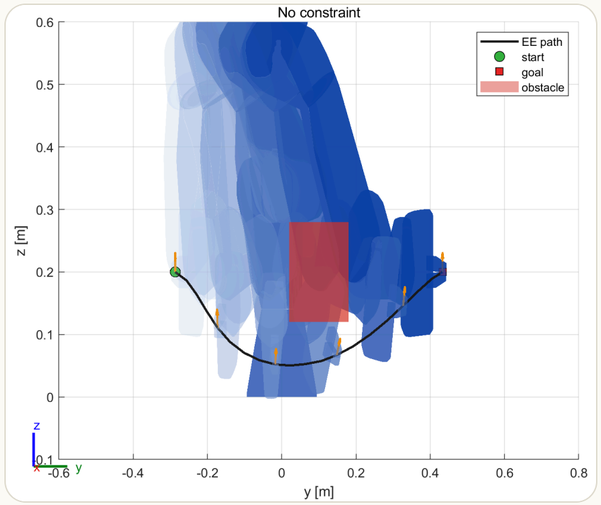
**19.1° → 0.8°**
mean tilt

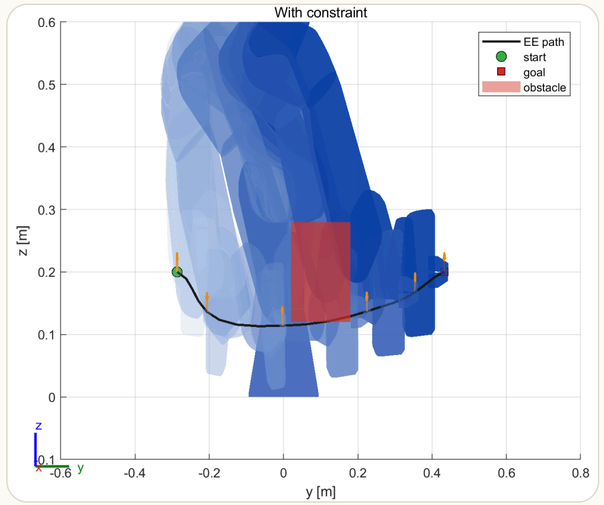
**27 → 124**
iterations; both collision-free

Orange arrow: the gripper's y-axis. It tilts on the left and stays vertical on the right; both plans avoid the box.

# Task 5: tilt at every waypoint



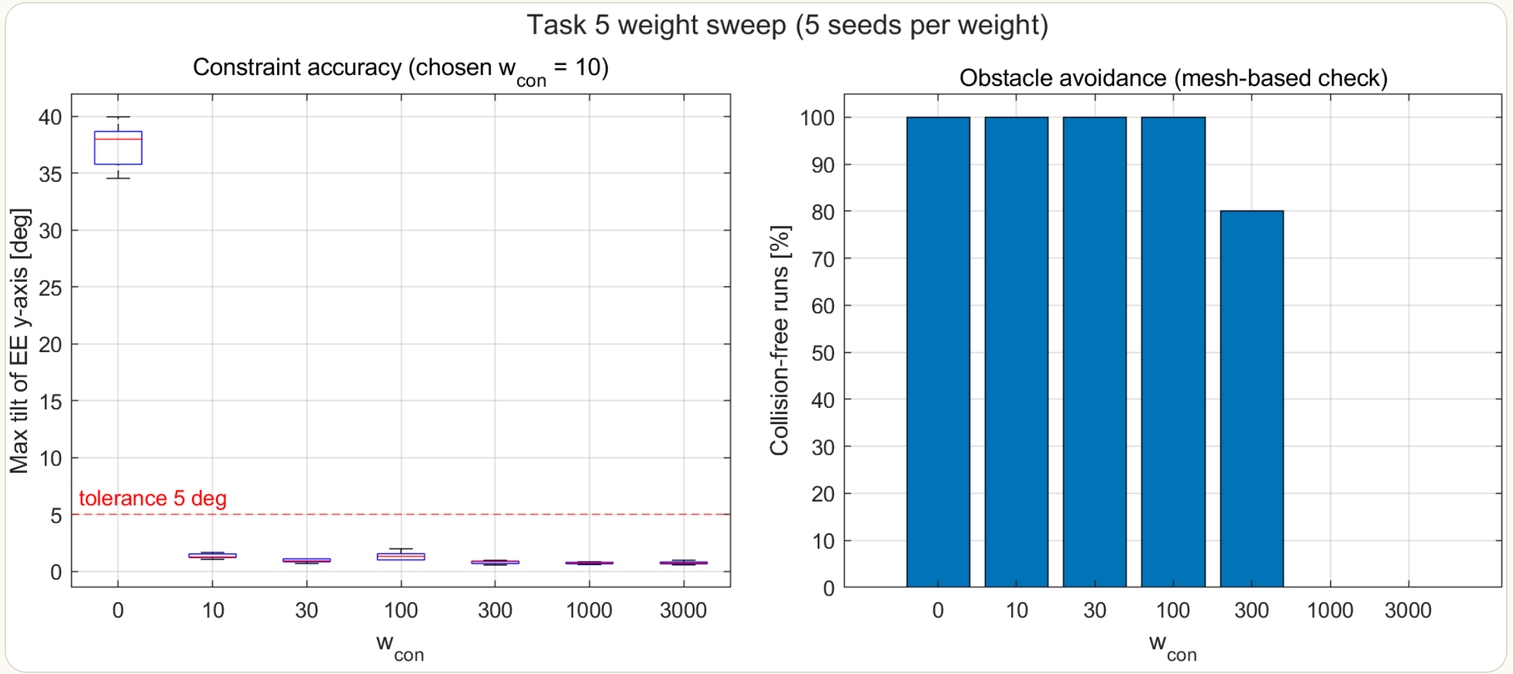




**Similar path length:** 0.839 m without, 0.821 m with the constraint.

**Same cost per iteration:** about 37 ms. The extra time comes from needing more iterations.

# Task 5: choosing the constraint weight




## w = 10

smallest weight that is collision-free on all 5 seeds, with tilt under 5°

### Too strong fails

At 1000 and above, no run is collision-free: the constraint overrides avoidance

# All five tasks solved and verified

## What we added

- Joint limits and sample reuse
- Spheres fitted to the meshes
- Between-waypoint collision checks
- 10 sample-code bugs fixed; a GUI

## Where the limits are

- Spheres hold 99% of the mesh, not all
- STOMP is local and depends on its start
- The upright constraint is soft: 1.9°, not 0°

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**Joshua:** Tasks 1–2, framework, patience · **Jiamo:** Tasks 3–4, extensions, GUI · **Dexter:** Task 5, plots

AI (Claude) was used to aid the code, report and presentation.

## Thank you. Questions?